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Hexagonal Boron Nitride-Based Electrolyte Composite for Li-Ion Battery Operation from Room Temperature to 150 °C

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Batteries for high temperature applications capable of withstanding over 60 °C are still dominated by primary cells. Conventional rechargeable energy storage technologies which have exceptional performance at ambient temperatures employ volatile electrolytes and soft separators, resulting in catastrophic failure under heat. A composite electrolyte/separator is reported that holds the key to extend the capability of Li-ion batteries to high temperatures. A stoichiometric mixture of hexagonal boron nitride, piperidinium-based ionic liquid, and a lithium salt is formulated, with ionic conductivity reaching 3 mS cm^{-1}, electrochemical stability up to 5 V and extended thermal stability. The composite is used in combination with conventional electrodes and demonstrates to be stable for over 600 cycles at 120 °C, with a total capacity fade of less than 3%. The ease of formulation along with superior thermal and electrochemical stability of this system extends the use of Li-ion chemistries to applications beyond consumer electronics and electric vehicles.

1. Introduction

Lithium-ion batteries (LIBs) present a combination of extended cycle life and high energy density, which made them the standard power source for portable devices. However, a long-standing technical challenge has been impeding its application even at mildly elevated temperatures (≈60–80 °C) without a significant loss on device performance.[1–7]

The most critical problems while operating LIBs at high temperatures arise due to the failure of separator and electrolyte. The former is usually composed of a thin polymeric film, which can soften or shrink upon heating, resulting in electrical short circuit.[2,3] Commercial electrolytes are based on organic solvents with low boiling points and so an increase in temperature elevates the internal pressure of the cell, potentially damaging the electrodes and the packaging. Besides being prone to failure, there are huge safety concerns with the cells on both situations, as exemplified by battery blown-up occurrences. Operation at high rates may lead to thermal dissipations that increase the internal temperature of the cell,[4,5] favoring the occurrence of runaway and failure. In this scenario, the development of temperature-friendly technologies is a vital step to meet the safety expectations for these widely used devices.[8,9]

In addition to being able to handle accidental exposition to high temperatures, the development of safe and reliable devices capable of working at temperatures over 100 °C would benefit several applications.[1,7] Early attempts on this regard involved the use of lithium-containing solutions in ionic liquids (RTIL) as the electrolyte, due to its high thermal stability and negligible vapor pressure at all temperatures before it decomposes.[10–13] Nevertheless, the mechanical stability of the separator is still an issue,[3] although there have been a few promising reports.[2] Polymer electrolytes are often regarded as good alternatives for high temperature electrochemistry, but they tend to lack mechanical robustness for long exposure to temperatures higher than 100 °C.[1,14]

We recently reported[15] a composite of natural bentonite clay and ionic liquid that can simultaneously act as electrolyte and separator, allowing a supercapacitor to work at temperatures as high as 200 °C. In the present work we report a lithiated electrolyte system with improved transport properties and thermal stability, allowing the fabrication of Li-ion batteries possessing good cycle life from room temperature to 150 °C.

2. Results and Discussion

Our previous efforts in developing electrolytes for supercapacitors[15] and Li-ion batteries[16] using a clay-based system were successful in showing the possibility of attaining long term electrochemical performance at extreme environments using ceramic–RTIL composites. A deeper understanding of the ionic transport mechanisms on this quasi-solid state system could be beneficial for tailoring similar compositions with improved performance in Li-ion battery applications. Bentonite clay is known to have a layered structure containing exchangeable ions and high charge density in the interlayer space.[17] The presence of

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additional ionic species other than Li^+ in the solid may hinder lithium ion transport, building up concentration overpotential on a cell level. Hexagonal boron nitride (h-BN) arises as an interesting counterpart to bentonite for use in ceramic composite electrolytes for LIB, as a model layered material with no permanent charges or movable cations in its structure, possessing very high thermal conductivity, being thermally stable and chemically inert.[18,19]

The electrolytes were prepared by mixing the ceramics with an appropriate weight of a 1 mol L^{-1} solution of LiTFSI in the ionic liquid 1-methyl-1-propylpiperidinium bis(trifluoromethylsulfonyl) imide (PP13). Although bentonite and hexagonal boron nitride have a similar flake size of ≈1 µm, h-BN was found to absorb twice as much RTIL solution as clay before it reached the same paste-like consistence, probably due to a higher specific volume of the powder. The ionic conductivities of RTIL solution, clay–RTIL and h-BN–RTIL electrolytes were recorded from room temperature to 150 °C, as shown in **Figure 1**a. The addition of ceramic particles was found to reduce ion mobility, with a more marked effect at lower temperatures. At 24 °C, the conductivity for the h-BN composite is ≈0.2 mS cm^{-1}, above the benchmark value of 0.1 mS cm^{-1} for battery electrolytes,[20] while the clay system lies below. The ionic conductivity dependence on temperature followed a Vogel–Tammann–Fulcher (VTF) behavior (Figure S1, Supporting Information) and the fitted parameters are presented in Table S1 in the Supporting Information. The pre-exponential factor A is proportional to the number of charge carriers in the electrolyte[21] and although it evidently dropped with addition of ceramic components, it is interesting to note that the ratio between $A_{h\text{-}BN}$ and A_{clay} is ≈2. Assuming that both composites have comparable densities, this factor is larger than the expected simply due to the presence of ≈30% more ionic liquid solution in the h-BN system, suggesting that the actual nature of ceramic within the electrolyte impacts the ion distribution and the effective free volume. Interestingly, the presence of ceramic particles resulted in a decrease in the pseudo-activation energy, possibly due to a local partial screening of coulombic interactions in the ionic liquid, with a more significant effect observed with the presence of the charged platelets of clay. The ideal glass transition temperature is higher in the composites than in the RTIL solution, indicating stronger solute–solvent effects.

Although the plots in Figure 1a depict the overall ionic motion in the investigated electrolytes, it does not represent the Li^+ conductivity, which is the essential metric for LIB operation. The ionic nature of RTILs imply in the existence of several charged species other than Li^+ migrating in the cell, which may deplete the concentration of lithium ions in the electrode vicinity. This would contribute to an increase in concentration overpotential and thereby affects device operation, especially at high rates. The lithium ion transference number (T_{Li+}) was measured following the ac/dc polarization method proposed

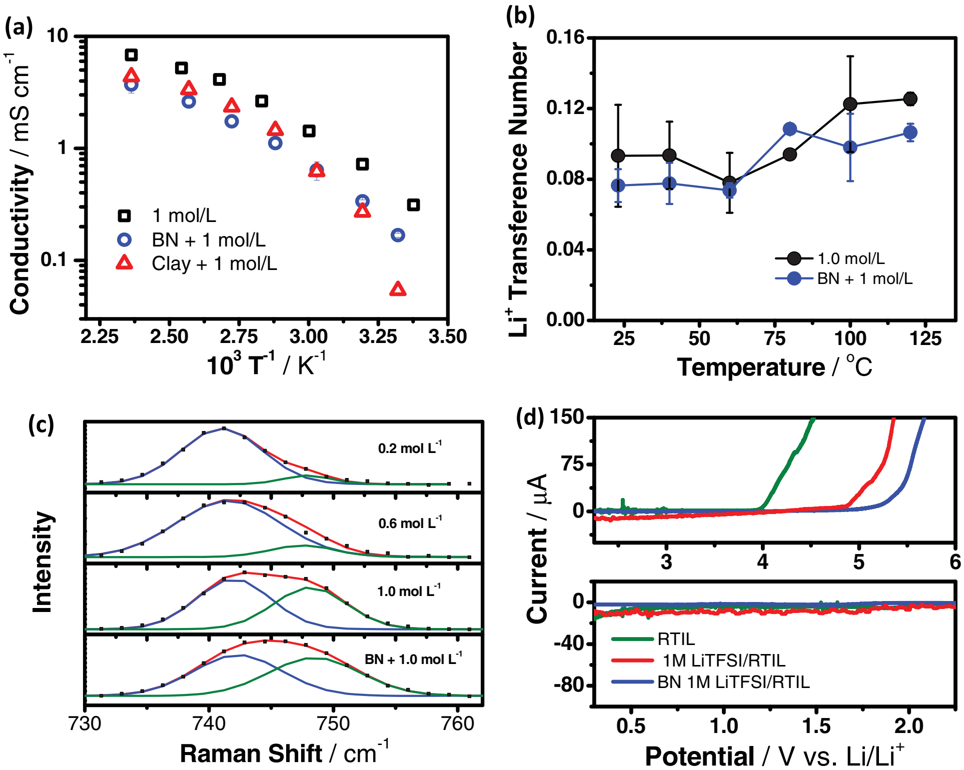

Figure 1. Measurement of transport and electrochemical properties of the electrolyte systems: a) Arrhenius plot of ionic conductivity for the 1 mol L^{-1} solution of LiTFSI in PP13 and for the BN- and clay-based composites; b) evolution of T_{Li+} with temperature for h-BN composite and for the LiTFSI/PP13 solution; c) Raman spectrum of the BN system in the region of $TFSI^-$ full anion vibration. The spectra for LiTFSI solutions in RTIL with different concentration are also provided for comparison. The dots are experimental points and the red line is the fit obtained combining the two band components; d) anodic (top) and cathodic (bottom) linear voltammetry scans at 120 °C for the neat ionic liquid, the 1 mol L^{-1} LiTFSI/PP13 solution and the h-BN-based electrolyte. The legends apply for both plots.

by Evans et al.[22] The technique is based on the assumption that in a symmetric cell containing electrodes which are non-blocking only towards lithium ions the application of a constant bias over a long time leads to a steady state, in which the overall current flow corresponds to the flux of lithium ions only. A typical polarization measurement is shown in Figure S2a in the Supporting Information, with the impedance plots before and after applying the bias shown as inset. The resistance of the passivation layer formed on top of the lithium metal electrodes was in the range of hundreds of ohms at room temperature, quickly dropping upon heating the cell (Figure S2a, Supporting Information).

The T_{Li+} at 23 °C was found to be 0.093 ± 0.029, 0.076 ± 0.009, and 0.060 ± 0.001 for the 1 mol L^{-1} solution of LiTFSI in PP13, the h-BN and the clay composite, respectively, and the values are in the same range of other reports for ionic liquids in the literature.[10,23,24] Changes in the transference number may occur as a consequence of adding extra ions to the electrolyte or by uneven changes in the diffusion constants of charge carriers. Hexagonal boron nitride is expected to weakly interact with ionic liquids[25] and it is not clear how its presence can affect T_{Li+}. Additionally, the negatively charged clay platelets may also influence the ion mobility in the electrolyte. Ceramic fillers at small concentrations have been widely reported to improve transport properties in polymer electrolytes by promoting salt dissociation[26] and avoiding chain recrystallization.[27] Nevertheless, at higher concentrations the particles tend to aggregate, reducing the effective contact area with the electrolyte and negatively affecting charge transport. Since the clay composite has a high solid content, the actual role of the ceramic particles might be reduced due to phase segregation. Another factor to consider is that ion exchange might be happening at a small extent between the liquid phase and the clay, thus depleting the concentration of Li^+ in the electrolyte and reducing the transference number. Due to the improved ionic conductivity and lithium ion mobility, the h-BN-based composite was further characterized for use in LIBs operating at a wide temperature range.

The h-BN composite comprises a uniform distribution of boron nitride particles and ionic liquid, as observed using scanning electron microscopy in Figure S3 in the Supporting Information. The specific interactions between the two phases are very weak and no major changes were observed in the Raman spectrum of the composite as compared to the 1 mol L^{-1} solution of LiTFSI in ionic liquid (full data not shown here). However, the solvation of lithium ions by $TFSI^-$ can be monitored by analyzing the band at ≈745 cm^{-1} assigned to the full anion vibration[28] as shown in Figure 1c. This band can be resolved into two components, originated from the vibration of free anion (≈742 cm^{-1}) and Li^+-bound $TFSI^-$ (748 cm^{-1}). The plots clearly show that addition of lithium salt increases the fraction of anions involved in strong solvation, as indicated by the enhanced intensity of the high frequency component. The analysis of the spectra also show that addition of hexagonal boron nitride increases the relative area of the bound $TFSI^-$ component by ≈5%, showing that Li^+ is actually more intensely solvated by the anions in the presence of the ceramic. Since the h-BN does not show strong interactions with the other components, this effect might be a consequence of a confinement of the ionic liquid solution by the h-BN particles, reducing the average separation between the ions in the system. This is also in agreement with the higher T_0 values found after fitting the ionic conductivity data using the VTF model (Figure S1, Supporting Information).

Temperature effects on lithium ion transference numbers were also measured for both the RTIL solution and the h-BN electrolyte (Figure 1b). The values present only a slight increase above 80 °C, indicating that the Li^+ solvation remains nearly unchanged within the temperature range and that the temperature dependence of the diffusion coefficients of all species in the liquid phase is about the same. At higher temperatures, the resistance of the passivation layer formed on top of the lithium electrodes in the symmetric cells drops drastically and the increased ionic mobility favors the achievement of a steady state much quickly than at room temperature (Figure S2b, Supporting Information).

The boron nitride–RTIL composite presented adequate conductivity and stable Li^+ transport properties in the range of 23 to at least 120 °C. However, the successful application in LIBs requires compatibility with an appropriate potential range, without the presence of unwanted reactions. The electrochemical stability of the electrolyte was evaluated using 3-electrode measurements, using lithium metal as both the counter and reference electrodes, with a stainless steel working electrode (see the Supporting Information for experimental details). Linear voltammetry scans taken at 120 °C are shown in Figure 1d for the RTIL, the 1 mol L^{-1} solution of LiTFSI in PP13 and for the h-BN composite (data at 23 °C is provided as Figure S4a, Supporting Information). Changes in the electrochemical stability of RTILs both with temperature increase and lithium salt addition have been reported by other groups,[29–31] but it still lacks a satisfactory explanation. It is clear from Figure 1d that the anodic stability of the RTIL at high temperature increases after addition of LiTFSI, with an even larger enhancement attained when h-BN is mixed with the RTIL-based solution. The electrochemical window was found to be broader than 4.7 V (from 5 down to <0.3 V vs Li/Li^+) at 120 °C, thus enabling its use with high energy density cathodes and anodes. Since the anodic stability might be strongly related to the reactivity of $TFSI^-$, the intensification of Li–TFSI interactions with addition of h-BN, as observed by Raman spectroscopy, might explain the observation of a wider electrochemical window. The electrolyte was also found to be compatible with lithium metal electrodes, showing a reversible plating/stripping behavior both at room and high temperatures (Figure S4b,c, Supporting Information). Additional discussion of the electrochemical window is provided as supporting information.

Although there are several reports on effects of cell overheating and abuse of lithium secondary batteries,[4,5,32–34] little is known about actual devices operating at temperatures above 80 °C and it is still necessary to understand thermal effects on fundamental cell processes. High energy density anode materials generally rely on the formation of a stable passivation layer to achieve stable performance and it is rather unpredictable how its complex and heterogeneous structure would behave if exposed to such conditions. In order to simplify the system on a device level, cell testing was performed using half-cells based on lithium titanate ($Li_4Ti_5O_{12}$, LTO), which presents a high

FULL PAPER

lithiation potential of ≈1.5 V and the absence of a thick, permanent solid-electrolyte interphase (SEI) layer.[35]

LTO half-cells containing the h-BN-based electrolyte were tested at 120 °C and electrochemical performance is shown in **Figure 2** and Figure S5 in the Supporting Information. The cyclic voltammograms (Figure 2a) show sharp and symmetric peaks corresponding to the lithium insertion and extraction in the electrodes, with the complete absence of additional reactions. Cells were tested by constant-current charge–discharge (CD) at different C-rates (Figure 2b and Figure S5a–c, Supporting Information), showing unprecedented cyclic stability, without significant capacity fade even after extended testing times. As an example, Figure 2b shows the cyclic stability for a cell cycled at a ≈C/8 rate, where the capacity is nearly constant at 158 mAh g^{-1} for more than 50 cycles and a total test

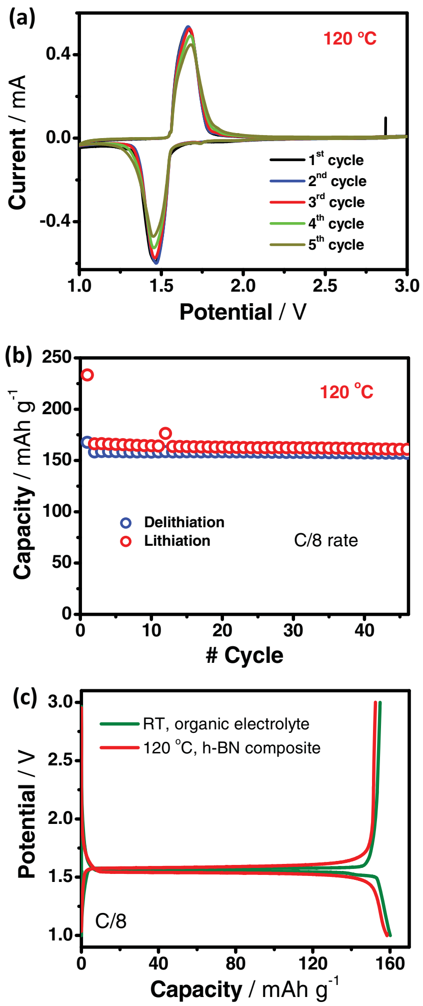

Figure 2. Performance of LTO half-cells containing the h-BN composite as electrolyte: a) cyclic voltammograms taken at 120 °C with a scan rate of 0.1 mV s^{-1}, showing symmetric and sharp peaks; b) cyclic stability for a cell operated for 32 days at a C/8 rate at 120 °C, showing stable capacity of 158 mAh g^{-1}; c) comparison between the charge–discharge profiles of a half-cell containing the h-BN composite (cycled at 120 °C) and a half-cell containing 1 mol L^{-1} $LiPF_6$ in EC:DMC 1:1 (v/v) (cycled at 23 °C), obtained at a C/8 rate. The performance is very similar, with both systems showing low polarization.

time of over a month. The lithiation kinetics was found to be so favored at high temperatures that changing the C-rate had only minor effects on the delivered capacity (Figure S5e, Supporting Information).The absence of thermal runway on the cells and the non-flammability and negligible vapor pressure of the electrolytes show that the h-BN composite allows reliable and safe operation for Li-ion batteries.

In contrast to previous reports in the literature of Li-ion cells operating within this same temperature,[1,7] the half-cells prepared using this composite electrolyte delivered exceptional coulombic efficiency values at 120 °C for a broad range of current densities even after several charge–discharge cycles, as presented in Figure S5d in the Supporting Information. In general, efficiencies over 98% were observed, showing the high reversibility of the electrode reactions at such extreme conditions and indicating that the electrolyte is stable under a long term exposure to harsh environments while involved in electrochemical processes. The high capacity values and low polarization observed for the cells at 120 °C were actually comparable to the performance of devices prepared with a conventional organic electrolyte and operated at room temperature (Figure 2c).

Since the electrolyte presented reasonable ionic conductivity and T_{Li+} at 24 °C it was also operational at room temperature. The sluggish transport properties at reduced temperatures accounted for a larger overpotential (Figures S6a,b, Supporting Information) and high total resistance (Figure S6c, Supporting Information), but stable capacities up to ≈90 mAh g^{-1} were achieved, over 50% of that at 120 °C. However, full capacity could be achieved at temperatures as low as 60 °C, as shown in Figure S5f in the Supporting Information.

A big challenge in successfully operating an electrochemical device at 120 °C is to assure that all components are stable over long periods under such conditions. Although most of the literature discusses the thermal stability of ionic liquids based on thermogravimetric analysis (TGA), this might not be a realistic figure of merit for applications that require long term exposition to high temperatures.[36,37] Chemical transformations in the ionic liquid at lower temperatures may form non-volatile electrochemically active species that can strongly affect cell performance. As an example, Meine et al.[38] found that alkylimidazoles were being formed after heating certain imidazolium-based RTILs for 24 h at temperatures above 120 °C.

To probe the long term stability of the electrolyte, aging tests were conducted by exposing a sealed sample of the h-BN composite to high temperature for different periods and then using it to prepare LTO half-cells. Exposition of the electrolyte paste to 120 °C for a period of 20 days did not lead to any visible changes (Figure S7a, Supporting Information). High temperature electrochemical performance of the cells prepared using such aged electrolyte was similar to the one presented by half-cells containing fresh electrolyte. Impedance spectra for cells using the composite aged for 10 and 20 days are presented in Figure S7b,c in the Supporting Information, respectively, and the only changes are a small increase in the ESR and the existence of a semicircle at high frequency. Small changes in the electrolyte conductivity can happen upon electrolyte aging,[39,40] while the development of a semicircle can be related to a small increase in the charge transfer resistance or due to the passivation of the lithium metal electrode. From the symmetric cells used to determine Li^+ transference number it is known that any passivation film would have a very low resistance at 120 °C (typically < 1 Ω) and so charge transfer may account for most of the change. Additionally, the cells were tested by cyclic voltammetry (CV) and the observed profiles for aged samples were found to closely resemble the one for cells using fresh composite (compare Figure S7d in the Supporting Information with Figure 2a). This further supports the superb stability of the electrolyte towards high temperature environments and limited effects of aging in the overall performance. Scanning electron microscopy images (Figure S3a–c, Supporting Information) also support that there are no major changes in the state of boron nitride particles in the electrolyte even after extensive cycling. Aging studies on the LTO electrodes and cycling effects on the half-cell impedance are provided as supporting information, along with a detailed discussion.

Although the electrolyte presents noticeable performance at 120 °C, high temperatures could possibly allow thermally activated processes to happen, triggering additional self-discharge mechanisms. Wang et al.[11] reported that some RTILs are incompatible with electrodes in the charged, high energy state, presenting an onset for thermal runway at temperatures lower than for conventional EC:DEC-based electrolytes. An alternative way of probing the stability of the h-BN composite electrolyte toward the charged electrode at high temperatures is through the analysis of self-discharge behavior and further stability of the cell upon recharge.

The profiles of the galvanostatic charge–discharge cycles before and after the self-discharge, along with the self-discharge curve itself, are shown in Figure S10a in the Supporting Information. It is possible to see that during the self-discharge (red portion of the plot) the cell potential quickly rises to a value slightly lower than the plateau voltage for LTO. The capacity retrieved from the cell after 24 h of open circuit at 120 °C corresponds to 97% of the cell capacity and no permanent capacity loss was observed after the self-discharge (Figure S10b, Supporting Information). The high extent of capacity retention and the absence of irreversible losses after self-discharge provide an independent evidence of the electrolyte stability in harsh environments, even in the presence of a charged electrode.

The overall cell performance obtained at 120 °C excels in all levels and draws a clear picture that the testing conditions were still far from the electrolyte possibilities. In order to show perspectives of this technology, we pushed the limits even further, by increasing the temperature of operation of the half-cell to 150 °C. **Figure 3**a shows a cyclic voltammetry at 150 °C for a LTO half-cell, showing that the electrochemical stability of the electrolyte holds up at higher temperatures and that the mechanical properties of the h-BN system are good enough to avoid short-circuits. Although the peaks display some extent of broadening after the first few cycles, the cell is able to deliver a stable capacity with a high coulombic efficiency of 97%. A change in the shape of the peaks was also observed in CV measurements with slower scan rates and seems to be essentially time-dependent. Since it is unrelated to cell cycling, we postulate that it might arise from changes in the electrode that affect the overall kinetics of the device, possibly involving redistribution of the PVdF binder (see the Supporting Information, Section 9). To the best of our knowledge this is the highest temperature of operation ever reported for a Li-ion battery that can also perform at room temperature. The cycling behavior at 150 °C may be further improved by using an appropriate binder and by employing appropriate electrode processing techniques.

Studying anode half-cells was remarkably useful to evaluate and understand the performance of the electrolyte in cells operating at high temperatures. Since practical devices are largely built in a full-cell configuration, it is also important to evaluate the electrolyte compatibility with the chemistry and voltage ranges of cathodes. Initial attempts with extra-lithiated manganese oxide ($Li_{1+x}Mn_{2-x}O_4$, LMO) as electrode explored here show promising results. As clear from the cyclic voltammograms and the charge–discharge profile for a LMO half-cell at 120 °C (**Figure 4**a,b respectively), this material presents two lithiation plateaus, with onsets at ≈3 and ≈4 V, with a total practical delithiation capacity of 146 mAh g^{-1}. The high extent of reversibility of the electrode reactions observed in the CV plot is a good indication that LMO is suitable for high temperature applications and is proof that the composite electrolyte can cater to cathodic potentials.

Having confirmed the compatibility of the electrolyte at both anodic and cathodic potentials using lithium half-cells, a preliminary attempt was also made to test the operability of the electrolyte in a full-cell configuration. Using LTO and $Li_{1+x}Mn_{2-x}O_4$ as electrodes, cells were tested with the h-BN composite electrolyte and also with conventional organic electrolytes. The cell configuration was balanced by considering only the high voltage plateau of the cathode. Cyclic voltammetry tests performed at 120 °C for the h-BN resulted in comparable profile as that of the organic electrolyte at room temperature (Figure S11a,b, Supporting Information) and the lack of spurious peaks at high temperatures indicates the compatibility of the electrolyte with the potential range of a full-cell operation. The cell with h-BN electrolyte tested at 120 °C delivered

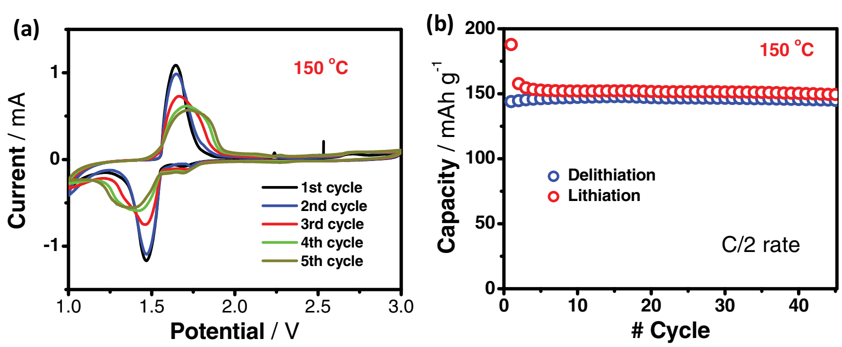

Figure 3. Perspectives for h-BN composite electrolyte: a) cyclic voltammetry and b) cyclic stability of a LTO half-cell containing the h-BN composite tested at 150 °C. Although there are changes in the electrode kinetics over time, the electrolyte is electrochemically stable and the capacity fade is negligible.

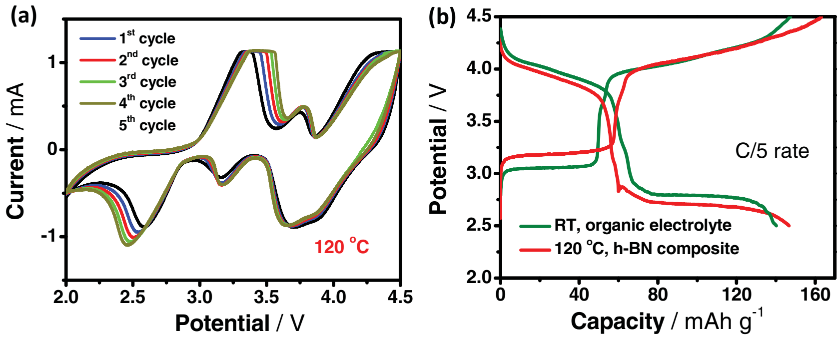

Figure 4. Compatibility of the h-BN electrolyte with cathode materials: a) cyclic voltammogram of a $Li_{1+x}Mn_{2-x}O_4$ half-cell tested at 120 °C; the charge–discharge profile in (b) shows that at high temperature the h-BN electrolyte allows performance similar to conventional electrolytes tested at room temperature.

a stable voltage profile with distinctive plateaus for both anodic and cathodic potentials, resulting in a reasonable cyclic stability (Figure S11c,d, Supporting Information). Issues such as cathode stability at high temperatures would need to be further probed for improving the cell performance and are not within the scope of this work. These initial results build a solid starting point for developing stable battery systems with high temperature stability.

3. Conclusions

We presented an electrolyte composite based on hexagonal boron nitride and ionic liquid that allows Li-ion battery operation with high efficiency over an unprecedented temperature range, spanning at least from 24 to 150 °C. The presence of a ceramic component in the electrolyte had little overall effect on the transport properties, although contributing to an increase of the electrochemical window. The composite was also shown to be compatible towards a cathode material, enabling full-cell testing at high temperatures. The development of practical battery technologies that enable cell operation over a wide temperature range has potential impact on a broad set of technologies, including defense, aerospace, and oil and gas applications.

4. Experimental Section

Electrode Preparation, Electrolyte Composition, and Cell Assembly: The detailed process is provided as Supporting Information. In short, electrodes were fabricated by coating slurries containing 80% active material, 10% graphite, and 10% poly(vinilydene difluoride) binder onto current collectors. The h-BN composite was prepared by mixing boron nitride powder and a 1 mol L^{-1} solution of LiTFSI in the RTIL 1-methyl-1-propylpiperidinium bis(trifluoromethylsulfonyl)imide, in a 1:2 w/w ratio.

Electrochemical Characterization: Galvanostatic CD tests were performed on a BT-2000 battery cycler (Arbin Instruments), while CV and electrochemical impedance spectroscopy (EIS) were performed on an Autolab PGSTAT 302N potentiostat. CVs were taken at 0.1 mV s^{-1} and EIS was performed at open circuit potential, in the range 1 MHz–100 mHz with amplitude of 10 mV.

Raman measurements were conducted by sealing the samples in glass tubes in a glove box and the data was collected in a backscattering geometry using a Renishaw spectrometer with a 514 nm laser and a 5× objective.

Ionic conductivities were measured by EIS, with the electrolyte placed between two stainless steel current collectors in a Swagelok cell without a spring, in a way to fix the cell constant at all temperatures. The electrolyte thickness was taken after the measurements were performed. For the RTIL solutions, conductivity measurements were taken using filter paper as separator.

All aging experiments were performed by sealing the cell component (electrode or electrolyte) in a coin cell case and storing it under the appropriate condition. The assembly and disassembly of cases were always performed in a glovebox. Lithium ion transference numbers (T_{Li+}) were measured using symmetric Li/electrolyte/Li cells. Experimental details for T_{Li+} calculation, electrochemical stability determination, and self-discharge are provided as supporting information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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[1] Q. Hu, S. Osswald, R. Daniel, Y. Zhu, S. Wesel, L. Ortiz, D. R. Sadoway, *J. Power Sources* **2011**, *196*, 5604.
[2] W. Jiang, Z. Liu, Q. Kong, J. Yao, C. Zhang, P. Han, G. Cui, *Solid State Ionics* **2013**, *232*, 44.
[3] H. Lee, M. Yanilmaz, O. Toprakci, K. Fu, X. Zhang, *Energy Environ. Sci.* **2014**, *7*, 3857.
[4] J. Shim, R. Kostecki, T. Richardson, X. Song, K. A. Striebel, *J. Power Sources* **2002**, *112*, 222.
[5] J. Kang, G. Rizzoni, *Int. J. Energy Res.* **2014**, *38*, 1994.
[6] S. S. Zhang, K. Xu, T. R. Jow, *J. Power Sources* **2006**, *154*, 276.
[7] L. Bodenes, R. Naturel, H. Martinez, R. Dedryvère, M. Menetrier, L. Croguennec, J.-P. Pérès, C. Tessier, F. Fischer, *J. Power Sources* **2013**, *236*, 265.
[8] P. G. Balakrishnan, R. Ramesh, T. Prem Kumar, *J. Power Sources* **2006**, *155*, 401.
[9] H. Maleki, G. Deng, A. Anani, J. Howard, *J. Electrochem. Soc.* **1999**, *146*, 3224.
[10] A. Lewandowski, A. widerska-Mocek, *J. Power Sources* **2009**, *194*, 601.
[11] Y. Wang, K. Zaghib, A. Guerfi, F. F. C. Bazito, R. M. Torresi, J. R. Dahn, *Electrochim. Acta* **2007**, *52*, 6346.
[12] S. F. Lux, M. Schmuck, S. Jeong, S. Passerini1, M. W. A. Balducci, *Int. J. Energy Res.* **2010**, *34*, 97.

[13] N. Byrne, P.C. Howlett, D.R. MacFarlane, M. Forsyth, *Adv. Mater.* **2005**, *17*, 2497.
[14] P. Novak, P. Podhajecky, *J. Power Sources* **1991**, *35*, 235.
[15] R. S. Borges, A. L. M. Reddy, M.-T. F. Rodrigues, H. Gullapalli, K. Balakrishnan, G. G. Silva, P. M. Ajayan, *Sci. Rep.* **2013**, *3*, 2572.
[16] K. Kalaga, M.-T. F. Rodrigues, H. Gullapalli, G. Babu, A. L. M Reddy, P. M. Ajayan, *ACS Appl. Mater. Interfaces* **2015**, *7*, 25777.
[17] J. Madejová, *Vibrational Spectrosc.* **2003**, *31*, 1.
[18] G. Gao, A. Mathkar, E. P. Martins, D. S. Galvão, D. Gao, P. A. S. Autreto, C. Sun, L. Cai, P. M. Ajayan, *J. Mater. Chem. A* **2014**, *2*, 3148.
[19] E. Husain, T. N. Narayanan, J. J. Taha-Tijerina, S. Vinod, R. Vajtai, P. M. Ajayan, *ACS Appl. Mater. Interfaces* **2013**, *5*, 4129.
[20] J. B. Goodenough, Y. Kim, *Chem. Mater.* **2010**, *22*, 587.
[21] M. Montanino, M. Moreno, M. Carewska, G. Maresca, E. Simonetti, R. Lo Presti, F. Alessandrini, G. B. Appetecchi, *J. Power Sources* **2014**, *269*, 608.
[22] J. Evans, C. A. Vincent, P. G. Bruce, *Polymer* **1987**, *28*, 2324.
[23] T. Fromling, M. Kunze, M. Schonhoff, J. Sundermeyer, B. Roling, *J. Phys. Chem. B* **2008**, *112*, 12985.
[24] C. Liu, X. Ma, F. Xu, L. Zheng, H. Zhang, W. Feng, X. Huang, M. Armand, J. Nie, H. Chen, Z. Zhou, *Electrochim. Acta* **2014**, *149*, 370.
[25] G. Kamath, G. A. Baker, *RSC Adv.* **2013**, *3*, 8197.
[26] F. Croce, G. B. Appetecchi, L. Persi, B. Scrosati, *Nature* **1998**, *394*, 456.
[27] H. Y. Sun, H.-J. Sohn, O. Yamamoto, Y. Takeda, N. Imanishi, *J. Electrochem. Soc.* **1999**, *146*, 1672.
[28] J.-C. Lassègues, J. Grondin, D. Talaga. *Phys. Chem. Chem. Phys.* **2006**, *8*, 5629.
[29] P. Reale, A. Fernicola, B. Scrosati, *J. Power Sources* **2009**, *194*, 182.
[30] V. Borgel, E. Markevich, D. Aurbach, G. Semrau, M. Schmidt, *J. Power Sources* **2009**, *189*, 331.
[31] J. Mun, Y. S. Jung, T. Yim, H. Y. Lee, H.-J. Kim, Y. G. Kim, S. M. Oh, *J. Power Sources* **2009**, *194*, 1068.
[32] G. G. Amatucci, C. N. Schmutz, A. Blyr, C. Sigala, A. S. Gozdz, D. Larcher, J. M. Tarascon, *J. Power Sources* **1997**, *69*, 11.
[33] K. Araki, N. Sato, *J. Power Sources* **2002**, *112*, 124.
[34] K. Amine, J. Liu, I. Belharouak, *Electrochem. Commun.* **2005**, *7*, 669.
[35] M.-S. Song, R.-H. Kim, S.-W. Baek, K.-S. Lee, K. Park, A. Benayad, *J. Mater. Chem. A* **2014**, *2*, 631.
[36] M. Kosmulski, J. Gustafsson, J. B. Rosenholm, *Thermochim. Acta* **2004**, *412*, 47.
[37] J. Salgado, M. Villanueva, J. J. Parajó, J. Fernández, *J. Chem. Thermodynamics* **2013**, *65*, 184.
[38] N. Meine, F. Benedito, R. Rinaldi, *Green Chem.* **2010**, *12*, 1711.
[39] N. Shubha, R. Prasanth, H. H. Hoon, M. Srinivasan, *Mater. Res. Bull.* **2013**, *48*, 526.
[40] Q. Li, T. Itoh, N. Imanishi, A. Hirano, Y. Takedaa, O. Yamamoto, *Solid State Ionics* **2003**, *159*, 97.